

Team MARTIAN - ISRO Hackathon PS13 Master Brief

What is this challenge?

ISRO wants a fully offline AI-powered Network Operations Center (NOC) assistant. The system should monitor a simulated MPLS/SD-WAN network, predict failures before they happen, explain the reasons behind the prediction, and recommend actions to prevent outages. Think of it as an AI network engineer that works without internet access.

How do we pull this off?

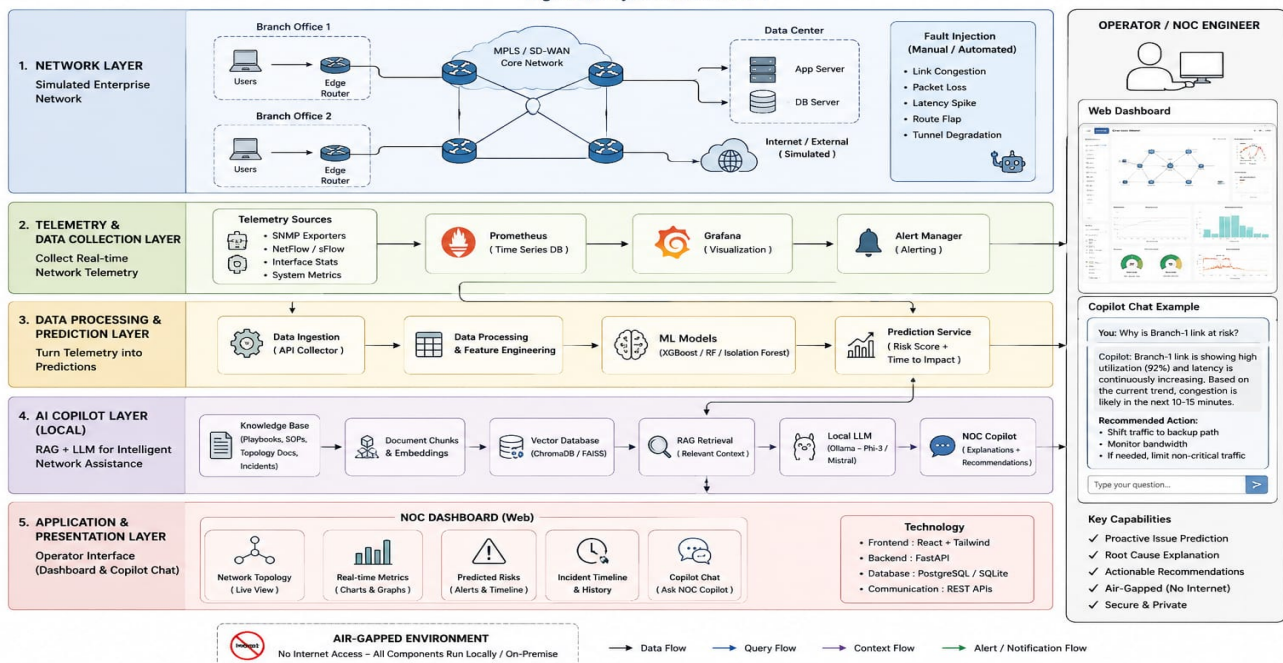
Step 1: Build a simulated enterprise network. Step 2: Generate traffic and intentionally inject faults. Step 3: Collect telemetry such as latency, packet loss, utilization, and routing events. Step 4: Train ML models to predict failures before they occur. Step 5: Connect predictions to a local LLM using RAG. Step 6: Present everything through a dashboard and copilot chat interface.

Why this project is valuable

This project teaches networking, AI, machine learning, observability, distributed systems, cybersecurity, dashboards, and real-world enterprise architecture. The skills gained are directly relevant to software engineering, DevOps, SRE, cloud engineering, AI engineering, and infrastructure roles. It is also one of the few ISRO challenges that can evolve into a practical product beyond the hackathon.

PS13 – AIR-GAPPED PREDICTIVE NOC COPILOT

High Level System Architecture



Project Roadmap

Week 1: Learn MPLS, BGP, OSPF and SD-WAN fundamentals. Week 2: Build network simulation using Containerlab and FRRouting. Week 3: Collect telemetry using Prometheus and Grafana. Week 4: Build prediction engine using XGBoost and anomaly detection. Week 5: Build local copilot using Ollama, Mistral/Phi-3 and ChromaDB. Week 6: Integrate dashboard, prediction engine and copilot.

Technology Stack

Network: Docker + Containerlab + FRRouting Traffic: iperf3 + ping + tc/netem Monitoring: Prometheus + Grafana Data: Python + Pandas + NumPy ML: Scikit-Learn + XGBoost Backend: FastAPI Copilot: Ollama + Phi-3/Mistral + ChromaDB Frontend: React + Tailwind + React Flow + Recharts Database: SQLite → PostgreSQL Deployment: Docker Compose

Five Knowledge Pillars

1. Networking 2. Telemetry & Monitoring 3. Predictive Analytics 4. LLMs & RAG 5. System Architecture & Integration

What Judges Care About

Prediction accuracy and lead time. Quality of copilot explanations. Air-gapped offline operation. Documentation and architecture clarity. A complete working system matters more than complex theory.

Team MARTIAN Mission

Build an air-gapped predictive NOC copilot capable of forecasting failures, explaining risks, identifying probable root causes, and helping operators take action before service degradation occurs.